Count significant digits

1 State the number of significant figures in the following numbers:

a $108\,486$ **b** 5105 **c** 0.20 **d** 10.20

e 84.00 **f** $0.007\,193\,0$ **g** $0.004\,58$ **h** 2.0058

2 State the smallest possible number of significant figures in the following numbers:

a $1\,500\,000$ **b** $26\,500$ **c** $1\,060\,000$ **d** $82\,420\,000$

3 A rainfall gauge has markings every $10\ \text{mL}$. If $369\ \text{mL}$ of rain fell in this gauge, to how many significant figures would the rain be recorded?

4 Beth just bought a house for $\$858\,400$. When asked for the price by her friend, she said she paid around $\$860\,000$. To how many significant figures did Beth round the price in her answer?

5 A number has been rounded to two significant figures. If the rounded number is 4500, find the largest possible value of the original number.

Round with significant figures

6 Consider the number 1083.

a Round 1083 to:

i One significant figure **ii** Two significant figures

b Find the difference between these two rounded values.

7 Round the following numbers to two significant figures:

a 2584 **b** $10\,694$ **c** 9249 **d** 215.99

e 0.7218 **f** 0.0297 **g** 4.95 **h** $0.006\,037\,736$

8 Round the following numbers to three significant figures:

a 1771 **b** $461\,585$ **c** 6484 **d** $20\,994$

e 9248 **f** 539.99 **g** 0.5628 **h** $0.073\,582$

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9 Round the following numbers to four significant figures:

a $801\,600\,713$	**b** $0.060\,070\,047$	**c** 11.8483	**d** $2.159\,76$
e $4.504\,13$	**f** $1.599\,64$	**g** 90.0335	**h** $0.598\,71$
i 41.5005	**j** 9.9996	**k** $0.093\,845$	**l** 808.5042

10 Round the following numbers to five significant figures:

a $5.659\,64$	**b** $5.600\,96$	**c** $12.006\,65$	**d** $39.999\,95$
e $7.034\,500$	**f** $0.041\,454\,9$	**g** $99.097\,906$	**h** $9.999\,99$

11 Round 7231.0108 to five significant figures.

12 Use your calculator to round the following to three significant figures:

$$\frac{(9.45)^9 + 8\pi}{\sqrt{\left(\frac{3}{2} + 5^2\right)}}$$

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